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Note: Electroplating process for connectorizing superconducting NbTi cables

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Experiments done at cryogenic temperatures below the 4.2 K temperature of liquid helium frequently require superconducting coaxial cables to efficiently transmit high-frequency signals while minimizing heat transfer to the experiment's cold stage. These cables are often made of niobium-titanium (NbTi) alloy which is difficult to solder directly. This note describes a new electroplating procedure for plating NbTi coax directly with copper, which enables connector attachment matching the structural and electrical properties of standard coaxial cables. Here, a cable is first electrochemically coated with a thin oxide layer and then electroplated with copper in an acidic copper sulfate solution. The procedure has modest safety requirements and may be carried out in a standard vented laboratory fume hood. *Published by AIP Publishing.* https://doi.org/10.1063/1.5030431

For cryogenic science and engineering experiments conducted at temperatures substantially below the temperature of helium liquid transition, 4.2 K, transmitting electrical signals between the cryogenic instrumentation and the supporting room temperature apparatus pose a well-known problem: how does one make a good electrical link without introducing excessive heat flux into the cryogenic system? An example of this general problem is our apparatus, designed to cool superconducting nanowire single-photon detectors (SNSPDs) to just below 0.8 K. Our commercial closed cycle cold head operating at ∼4 K has a substantial cooling power of about 130 mW.[1] Here, the "cooling power" is a measure of the heat load a refrigerating device can tolerate before its operating temperature is substantially increased above its ultimate minimum. On the other hand, the ^{4}He cryocooler that lowers the temperature to ∼0.8 K is rated for only about 100 μW of cooling power.[2] The problem is even worse for the dilution refrigerator systems used in most cryogenic quantum computing applications for superconducting qubits. For these, the available cooling power scales as $\sim T^2$, and for a typical system operating at 10 s of mK, the available cooling power is only a few μW.[3] Additionally, specific heats scale as a power law of the temperature.[4]

Even if the refrigeration system is capable of handling the heat load, an electrical link may also open up a large thermal link, causing a locally significant temperature increase of the components being studied. The core of this relationship between electrical conductivity and thermal conductivity is described by the Wiedemann–Franz law for regular metals, which implies that good electrical conductors will necessarily have large electronic thermal conductivity. Estimates show that the metal wires frequently used at higher temperatures will transfer an excessive heat load when bridging the sub-Kelvin and liquid helium–temperature segments.

The solution to this problem is well known: use superconducting conductors, which do not follow the Wiedemann–Franz law. For temperatures well below the superconducting transition, electronic thermal conductivity goes as a power of the temperature and, for very low temperatures, is predicted to fall-off exponentially.[5] Superconducting cables have been developed for this purpose,[6] and coaxial superconducting cables for high-frequency transmission are available commercially.[7] NbTi is the preferred superconductor for these cables because it has a high superconducting transition temperature (for a low temperature superconductor) of ∼10 K. However, these cables are mediocre electrical conductors *above* the superconducting transition, and a conventional metal cable is usually used for the segment of the electrical link that goes from the ∼4 K stage of the cryostat to the room-temperature environment.

A remaining difficulty facing the experimentalist that we treat here is to develop a process for attaching radio frequency (RF) connectors to a superconducting cable. There are two ways that regular cables are connectorized: mechanically joining (crimping) the connector onto the cable conductors or, alternatively, making solder connections. Unfortunately, conventional crimp-type RF connectors do not allow for good electrical connections to superconducting coaxial cables. Any connection to the cable's shield is especially unreliable. However, special crimp-type connections have been developed in which a series of copper sleeves are mechanically swaged onto the coaxial conductors, and these in turn may be used to attach connectors as for conventional cables.[8] This process, when done properly, allows for reliable connectorizing, but producing and attaching suitable sleeves is a highly specialized activity. As such, cables produced in this manner are expensive and have long lead times. The alternative, directly soldering the coaxial conductors to a connector, fails because the metals used in the cable have an extremely tough oxide layer that is impervious to even the most aggressive fluxes. A third potential approach for joining NbTi to other materials could involve spot welding

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(resistance welding) or ultrasonic welding techniques.[9] With proper equipment, these may provide another way to connectorize cables.

An alternative for connectorization is electroplating onto the cable a surface that *can* be soldered. One approach that is used commercially is electroplating nickel onto NbTi cables, using procedures similar to that described in a 1961 patent by Saubestre and Bulan[10] for plating niobium and its alloys. However, this technique has two significant drawbacks. First, nickel is ferromagnetic, which is a potential problem for experiments sensitive to the magnetic environment including some cryogenic qubit proposals. Second, to promote strong adhesion of the nickel plating to the niobium, the Saubestre and Bulan procedure uses a high temperature bake that is well above the 327 °C melting point of polytetrafluoroethylene (PTFE).[11] As PTFE is the dielectric usually used for NbTi cables, this is a fatal drawback for using this as a connectorizing process.

We investigated plating the cables directly with copper, which is only weakly diamagnetic and is readily solderable. Existing recipes for electroplating titanium and its alloys with copper use a solution of sodium dichromate ($Na_2Cr_2O_7$) and highly concentrated hydrofluoric acid (HF).[12] Sodium dichromate is an OSHA-regulated carcinogen and is not allowed in our local materials science laboratory/cleanroom facility.[13]

To overcome these obstacles, we developed a new electroplating procedure for directly electroplating copper onto NbTi cables. It uses comparatively safe chemicals and may be carried out in a standard vented laboratory fume hood, and it does not require a high temperature bake.

The procedure uses two electrolytic cells. One holds a basic oxidizing solution and the other holds an acidic copper sulfate solution that does the actual plating. Initially, the NbTi surfaces must be thoroughly cleaned and degreased, preferably in an ultrasonic cleaner. Then, the first electrolytic cell is used to deposit a special oxide layer on the NbTi that will allow the copper to adhere to the titanium. This bath consists of 110 g/l of sodium hydroxide (NaOH) in deionized water. The NbTi substrate is wired as the anode in the solution with a copper sheet as the cathode with a typical spacing of 5 cm. 5 V is applied to this circuit for 15 s and then 10 V for approximately 30 s. This stepped up voltage should be applied until the substrate darkens noticeably from the gray of the NbTi to a yellow or tan color, as shown in Fig. 1(a). The oxide layer produces a structural color.[14] We found that a very thin oxide layer of <35 nm leads to the strongest adhesion.

After rinsing in deionized water, the substrate is quickly transferred to the second electrolytic cell. This bath is an acidic copper plating solution made of 220 g/l of copper sulfate pentahydrate ($CuSO_4 \cdot 5H_2O$) with 40 ml/l of sulfuric acid (H_2SO_4) in deionized water. The NbTi is connected as the cathode and plated for 120 s (or as necessary to achieve the desired thickness of copper) at 0.032 A/cm^2, producing the uniform coating of copper shown in Fig. 1(b). For the copper electrode, oxygen-free copper should be used to limit impurities and increase the lifetime of the $CuSO_4 \cdot 5H_2O$ solution. For large or continuous operations, the plating solution should be continuously filtered and regularly replaced.

This procedure requires standard chemical safety procedures and protective equipment including gloves, a lab coat, and chemical safety goggles. All steps can be performed in a standard laboratory fume hood. Neoprene gloves should be worn when handling concentrated H_2SO_4 and NaOH.

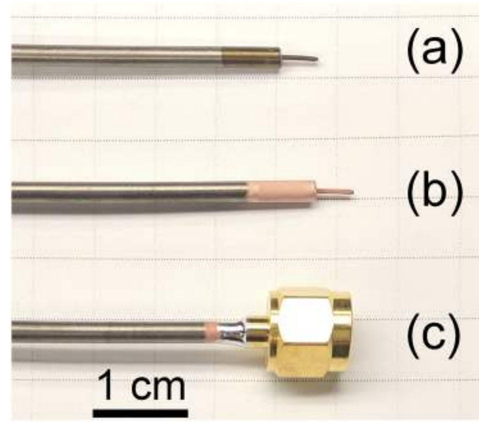

FIG. 1. (a) Oxidized NbTi cable. (b) NbTi cable electroplated with copper that is ready to be connectorized. (c) Electroplated NbTi cable with a soldered SMA (SubMiniature version A) connector.

For our purposes, we then attach SMA type connectors to the cables by soldering them just as if it was a standard coaxial cable.[15] A cable with an SMA connector attached is shown in Fig. 1(c). After producing each cable, we test them for thermal cycling by thermally shocking them in liquid nitrogen and warming them to room temperature. We repeat this cycle twenty times over the course of about 10 min. After cycling, the cables are tested for structural integrity and electrical conductivity. To date, none of the connectorized cables have failed.

We also have made several measurements on these cables to better understand the electrical and mechanical properties of the electroplated joints. The simplest of these measurements is to measure the contact resistance between the outer conductor and the connector at the point of contact, following the military's specifications for conducting four-wire resistance measurements on coaxial RF connectors (section 4.6.13 of MIL-PRF-39012E[16]). Due to the high resistance of the superconductor at room temperature, we simply place an upper bound on the outer conductor contact resistance of 1 mΩ, similar to that of SMA connectors attached to standard cables.[17]

Next, we conducted "pull tests" to characterize the mechanical integrity of the connector–cable junction. We carried out retention force tests on four cables by pulling on the SMA connector in a controlled, measured fashion until the connector broke free from the cable. Again following the military's specifications for the regular room temperature version of these particular RF cables (section 4.6.21 of MIL-PRF-39012E[16]), we found that the cable–connector junctions withstand 60 pounds of axial force. In the four tests we carried out, the junctions failed at 61, 67, and 69 pounds of force. We conclude that these cables are mechanically robust and may be handled just like similar cables using conventional conductors.

Finally, we carried out a test of the electrical properties of the connectorized cables by sending short pulses through a "U" made of two superconducting cables in the coldest stage of the cryostat. Signals traveled to and from the superconducting

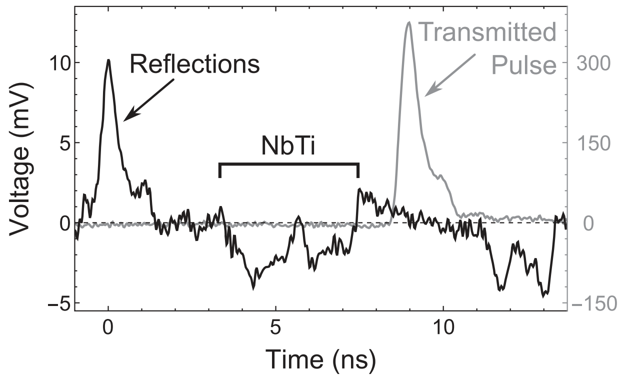

FIG. 2. Reflection of an electrical pulse sent through a cryostat containing two sections of the electroplated NbTi cable. The transmitted pulse height is 0.38 V.

segment along standard coaxial cables. All cables and connectors had a nominal characteristic impedance of 50 Ω. We sent a ∼700 ps full width at half maximum (FWHM) pulse—representative of signals in our SNSPD application—through the circuit and recorded on an oscilloscope the transmitted and reflected signals. The latter was extracted by using an RF circulator on the incident leg of the loop before it entered the cryostat, which would redirect any reflections coming back out of the chamber into the oscilloscope.

By looking at the time dependence of the reflected signal, we can assign different reflections to particular segments of the loop. The results are shown in Fig. 2. In effect, the loop, including the superconducting cables at the link, faithfully transmitted the incident signal. The reflection coefficient from the superconducting segment was <1%, corresponding to a reflection loss of better than −40 dB. Indeed, the connectorized superconducting cable was better impedance matched than other parts of the loop. Specifically, the first peak in the reflection curve of Fig. 2 is from a standard 50 Ω SMA vacuum feedthrough and is three times as large as the reflections from the connectorized cables.

In conclusion, we describe a new procedure for connectorizing superconducting cables and demonstrate the mechanical and electrical integrity of the joints. While our direct experience is with SMA connectors, this method should be readily adaptable to connecting any type of RF connector to any NbTi conductor. Finally, the copper on NbTi electroplating recipe we detail may have uses beyond our interest in attaching RF connectors to superconducting cables.

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